



Durability of flexurally strengthened RC beams with prestressed CFRP sheet under wet–dry cycling in a chloride-containing environment

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ABSTRACT

Using prestressed carbon fibre reinforced polymer (CFRP) to strengthen reinforced concrete (RC) members is currently considered a promising technology. This study aims to investigate the durability of RC beams externally strengthened with prestressed CFRP sheet in a chloride-containing environment. The prestressed load levels were set at 7.5%, and 15% of the ultimate tensile strength of the CFRP sheet at room temperature (~26 °C), and the test parameters included an inclined U-jacket (or not) and 90-day exposure to a wet–dry cyclic environment. The wet–dry cyclic condition was set as 8-h immersion in 3.5% NaCl solution at 40 °C and 16-h drying at 25 °C and 60% RH (relative humidity). After exposure to the wet–dry environment, the prestressed CFRP sheet strengthened RC beams were tested under four-point bending test. The evolution of time-dependent prestress losses in the strengthened beam was discussed, and then the flexural behaviour of the strengthened beams was analyzed. Finally, a theoretical model based on the classical beam theory was proposed to predict the bearing capacity of prestressed-CFRP strengthened RC beams. The results show that the prestressed CFRP sheet can significantly improve the flexural performance of RC beams, and the cracking and ultimate loads of the RC beam strengthened with a prestressing level of 15% could increase by approximately 50% and 40%, respectively. Moreover, the inclined U-jacket enhanced the flexural performance of the strengthened RC beams and changed the failure mode from CFRP debonding to CFRP fracture. Although the exposure in chloride-containing environments increased the prestress losses, a 90-day exposure in the simulated subtropical marine climate environment did not cause a significant detrimental influence on the flexural performance of the strengthened RC beams. The proposed model agrees well with the experimental results, indicating it could conservatively predict the bearing capacity of strengthened RC beams failed by CFRP fracture.

1. Introduction

Reinforced concrete (RC) structures employed within the marine environment are required to both bear loads and undergo cumulative damage caused by chloride ions [1]; therefore, the main problems threatening the service life of such RC structures are the combined effect of loading and corrosion caused by chloride ions [2], e.g. coastal bridge structures, the damage is shown in Fig. 1. Strengthening is the usual method employed to extend the structural life of these structures and to reduce maintenance costs [2,4]. In this respect, fibre reinforced polymers (FRPs) are widely-used in the repair and rehabilitation of existing damaged structures, and carbon FRP (CFRP) is well received

due to its specific strength, light weight, and chloride ion resistance, externally bonded CFRP sheets strengthening is shown in Fig. 1 [4–7].

Earlier studies conducted on the flexure performance of FRP-strengthened RC beams determined that the tensile strength of FRP was only 20–30% when the reinforcement yielding; and it was thus found to be difficult to restrain the development of structural deformation and cracks [8]. Therefore, FRP strengthening was proven to provide only a limited improvement in the flexural performance of the RC beams [9]. To increase FRP utilisation and achieve improved reinforcement effects, prestressed FRP-strengthening was proposed and developed [7,10]. A detailed review of the prestressing technique of FRP has been provided by Aslam et al. [8], and the full scope of that review will not be repeated here. As well known, near surface

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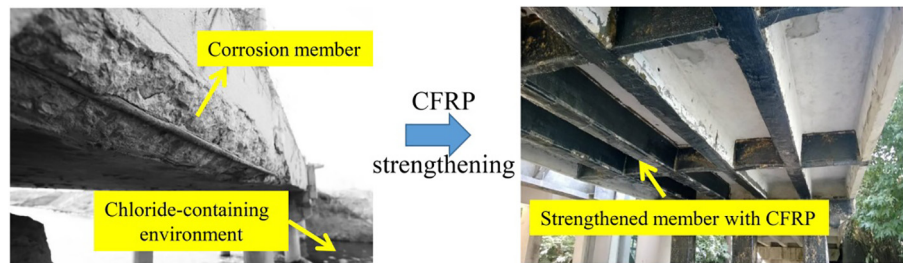


Fig. 1. Corrosion of RC member and CFRP strengthening.

mounted (NSM) FRP and externally bonded FRP are common techniques to rehabilitate existing structures [7,11,11,13]. Compared with prestressed NSM FRP, the strengthening technique of externally bonded FRP is easier to implement, although the NSM system could provide better load carrying capacity due to higher bond strength at the concrete-FRP interface [11]. As for prestressed externally bonded FRP, it has been found that the adhesive between the concrete substrate and FRP does not provide an adequate bonding capacity to resist the normal and shear stresses caused by the relatively high initial stress of the FRP laminate, and consequently debonding failure has been typically observed in externally bonded FRP-strengthened flexural members [11,13]. Therefore, anchorage systems for prestressed FRP have been developed [2,14]. The most frequently used forms are the mechanical [15], inclined U-jacket and gradient end-anchorage systems, and these methods have been shown to effectively improve the performance of prestressed CFRP-strengthened RC structures [12,16].

As well known, the short-term mechanical performance of RC structures could be improved by using FRP-strengthening technique [11,17]. However, relatively limited information is available for the durability of externally bonded FRP-strengthening system [3,5,19,19], particularly for the coupling effects of environment and sustained loading [19–23]. Kafodya et al. [20] studied durability of CFRP plates immersed in water and seawater under sustained bending, and the results indicated that immersion medium of distilled water and seawater and the degree of the bending strain by 50% did not show remarkable effects on the variation of the mechanical properties of CFRP. The increased sustained load may lead to variations in the pore structure of the resin matrix and the initial development of bond microcracks, and it also changes the permeation path [20,23]. In addition to conducting studies on the durability of FRP materials, Wei et al. [19] studied the durability of the interface between basalt FRP and concrete under the wetting and drying cycles in a chloride environment. The results indicated that the ductility and residual strength of the bond behaviour decreased with an increase in the wetting and drying cycles. Furthermore, Wang et al. [24] experimentally investigated FRP-concrete interface behaviour under wet-dry cycling, and found that wet-dry cycling can weaken the bond behaviour of CFRP-concrete.

As regarding to the durability of CFRP-strengthened concrete structure under marine environment, limited study has been reported. Tang et al. [25] studied the performance of FRP strengthened RC beam under wet dry cycling and a sustained load, the results indicated that combined wet-dry cycling and sustained loading had an adverse effect on the yielding and debonding load of the FRP-strengthened RC beams, but the U-shaped anchorages could reduce this influence. It is noted that very limited information is available for the durability of RC structures externally strengthened with prestressed CFRP laminate.

To this end, this study aims to investigate the durability of RC beams strengthened by prestressing CFRP sheet with an inclined U-jacket and the constituent materials (i.e. concrete, CFRP material, adhesive resin, and steel) under a wet-dry cyclic environment that

contains the chloride ion. The prestressing levels examined were set at 7.5%, and 15% of the ultimate tensile strength of the CFRP sheet at room temperature, and the inclined U-jacket was used to compare the reinforcement efficiency of the CFRP sheet. The evolution of time-dependent prestress loss in the strengthened RC beams was examined, and then the flexural behaviour of the strengthened RC beams after exposure was discussed. Finally, based on the consideration of the degradation of the constituent materials in the strengthening system caused by the exposure environment, a theoretical analysis was conducted to predict the bearing capacity of RC beams strengthened with prestressed CFRP.

2. Experimental program

2.1. Raw materials

The carbon fibre sheet (UT70-30) used in this study had a nominal thickness of 0.167 mm and width of 100 mm and was provided by Toray, Japan. To gain the tensile properties of CFRP sheet before and after exposure to the chloride environment, two sets of specimens (7 same specimens in each set) were prepared and tested in accordance with the guideline ASTM D3039 [26]. The adhesive adopted a two-component resin (HM-E8, Heroman Ltd., USA) with the base and hardener mixed at a ratio of 2:1 by weight. Similarly, two sets of dog-bone-shaped resin specimens (7 same specimens in each set) for the uniaxial tensile test were prepared following ASTM D 638 [27]. Fig. 2 illustrates the CFRP coupon and the dog-bone-shaped specimen for the adhesive. After curing 7 days at room temperature, the tensile strength, elastic modulus, and elongation at break of CFRP and adhesive resin were determined as 4132 MPa, 231 GPa, and 1.78%, 48 MPa, 2.7 GPa and 1.70%, respectively.

The commercial concrete from local supplier was used. The mix proportion of the concrete with a maximum aggregate size of 25 mm was designed according to the Chinese standard (JGJ 55-2011) [28]. The compressive strength and elastic modulus of cylinder concrete with 150 mm × 300 mm (diameter × height) were 35 MPa and 32 GPa after 28 days of standard curing, respectively. Furthermore, the yield strength, ultimate strength and elastic modulus of the tensile steel bars were 411 MPa, 625 MPa and 206 GPa, respectively.

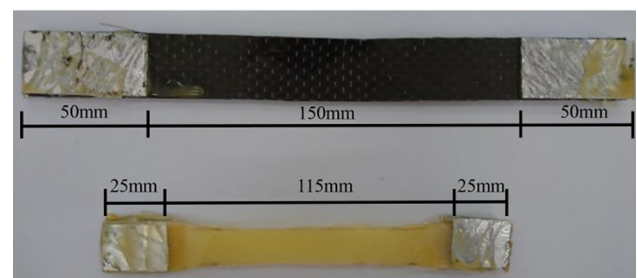


Fig. 2. Tensile specimens of CFRP and epoxy resin.

2.2. Test setup and durability test

The test beam measured 2300 mm (length) $\times$ 150 mm (width) $\times$ 200 mm (height) and span of 1800 mm. Longitudinal steel bars and erect bars were set as 2 Φ 10, and stirrups were set as 8@200. The reinforcement ratio of beams was designed to be 0.63%, which was higher than the minimum reinforcement ratio of 0.22%, to ensure these specimens were balanced-reinforced beam. Details relating to the specific sizes and reinforcement information are illustrated in Fig. 3.

One-layer carbon fibre sheet was used except control RC beams. The test parameters were set as follows: the prestressing level of CFRP sheet was set as 7.5%, and 15% of its ultimate tensile strength at room temperature, the inclined U-jacket, and corrosion time were set as 90 days. The design of test parameters and specimen numbers are shown in Table 1.

The timetable of the durability test of constituent materials and the strengthened RC beams in this study is shown in Fig. 4.

2.3. Pretensioning and strengthening procedure

The carbon fibre sheet was tensioned using a tensioning device designed by our team, which consists of a lifting counterforce table, a circular tooth anchor at the tensioning end, a fixed end, a roll shaft, anchor bolts, a tensioning screw, a baffle, a hydraulic jack and a force sensor (as illustrated in Fig. 5). The lifting counterforce table was used to ensure that the carbon fibre sheet closely connected with the specimen after tension. The friction between the carbon fibre sheet and the anchorage was enhanced using the circular tooth anchor and roller shaft. The friction was then further improved using the bolt constraint at the upper and lower parts of the anchorage, which reduced the prestressing loss between the carbon fibre sheet and the anchorage. Loading was applied using the hydraulic jack, which was connected to a tension screw, and force was monitored using the force sensor to ensure experimental accuracy. The maximum load of force sensor was 98 kN, the precision was 0.001 kN and the sensitivity coefficient was 1.4518 mV/V.

One-layer carbon fibre sheet (with a width of 100 mm) was used in the test. Prior to tensioning, the bonding surface of the RC beam was uniformly polished using an angle grinder until coarse aggregate was visible, and the polished surface was then cleaned with acetone. The adhesive was then coated on the surface of the carbon fibre sheet, and a roller was employed to ensure that the adhesive permeated evenly into the specimens and that no bubbles or defects occurred. The thickness of the adhesive layer was approximately 0.15 mm. The carbon fibre sheet was pasted on the RC beam before the adhesive cured. After a second roll pressing, weights were applied on the surface to prevent defects developing with respect to bulging. After eight days, the prestressed carbon fibre sheet was finally released for post curing in a laboratory environment ($\sim 26^\circ\text{C}$).

To study the effect of end anchorage, inclined U-jackets were bonded onto the prestressed carbon fibre sheet at an angle of 45° , which followed the suggestion of FRP-anchor angle by Fu et al. [3]. The U-jacket had a length of 460 mm and width of 100 mm (details of the U-jacket are illustrated in Fig. 6). Two LVDTs (YHD-100) were arranged in the middle and supports for monitoring the displacement,

the range of LVDTs was 0–100 mm and the precision was ± 0.001 mm.

2.4. Exposure environment

The durability test was conducted using a wet–dry cyclic test system (as illustrated in Fig. 7a) at the structural laboratory of Guangdong University of Technology, and the temperature of the laboratory is about 26°C . In order to simulate the high temperature and humidity marine environment in subtropical area of South China (the average annual temperature is 13°C to 20°C and the humidity is about 70%), the chloride ion concentration was 3.5%, and the wet–dry cycle occurred as follows: 40°C NaCl solution for 8 h and then drying at 25°C with 60% RH for 16 h. The temperature control precision was $\pm 2^\circ\text{C}$, and a sketched diagram of the cyclic process is illustrated in Fig. 7b.

2.5. Test procedure

In order to measure the time-dependent prestress losses of CFRP, eight strain gauges were attached on the surface of CFRP, as shown in Fig. 8. The strain gauges were in the dimension of 3 mm (width) $\times$ 5 mm (length), the resistance was 120 Ω , and the sensitivity coefficient is about 2.0%. The monitoring system including sensor (see Fig. 5) and strain collection instrument (TDS-530), and in the meanwhile the acquisition system (DT85G) was used to collect strains data.

The tests were conducted at the structural laboratory of Guangdong University of Technology. An electro-hydraulic testing machine (SDS-500 model, Changchun Academy of Mechanical Science Research Institute Co., Ltd., Changchun, China) equipped with a 1000 kN load cell was used to conduct the four-point bending test. Tensile tests for steel bar, CFRP and resin specimens were conducted using a Material Testing System instrument (MTS Systems, Eden Prairie, Minnesota, USA). The compressive performance of the concrete cylinders was tested on an Italian MATEST C088-O1 machine with a capacity of 4000 kN.

A TDS-530 high-speed static data logger (Tokyo Sokki Kenkyujo Co., Ltd, Tokyo, Japan) was employed to record the strain and deflection data. A crosshead speed of 1 mm/min under the displacement control was used in the bending test. The bending load was applied at 5 kN per grade during the initial stage, 10 kN per grade after concrete cracking, and then again at 5 kN at the point of near failure load. Crack development and deflection and strain were recorded following each per grade loading. The distribution of the strain gauges, LVDTs, and the loading method are illustrated in Fig. 6b.

3. Results and discussion

3.1. Durability of the constituent materials

In order to investigate the effect of chloride environment on the strengthening system, the CFRP sheet, adhesive resin, concrete cylinder and bare steel bar were exposed to the same corrosion environment as the strengthened RC beams. The influence of wet–dry cycling under the chloride-containing environment on the mechanical properties of the constituent materials is shown in Fig. 9. After 90

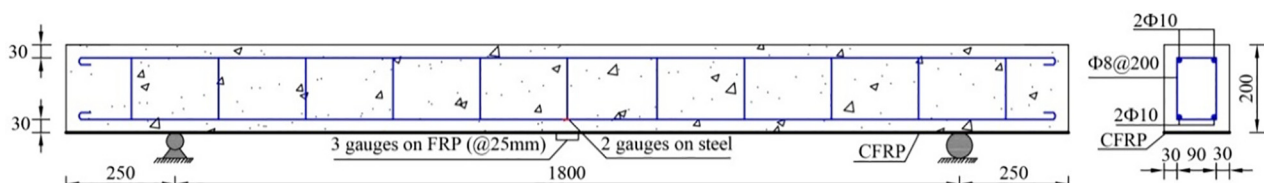


Fig. 3. Details of specimens (Units: mm).

Table 1
Experimental parameters.

Characteristics	^a Specimens	^b Prestress level	Inclined U-jacket	Corrosion time (days)
Control	RC	None	None	0
Effects of prestress level	SRC-0-0	0%	None	0
	SRC-7.5-0	7.5%	None	0
	SRC-15-0	15%	None	0
	SRC-0-90	0%	None	90
	SRC-15-90	15%	None	90
Effects of immersion time	SRC-0-0	0%	None	0
	SRC-0-90	0%	None	90
	SRC-15-0	15%	None	0
	SRC-15-90	15%	None	90
Effects of inclined U-jacket	SRCU-0-0	0%	Yes	0
	SRCU-7.5-0	7.5%	None	0
	SRCU-7.5-0	7.5%	Yes	0
	SRC-15-0	15%	None	0
	SRCU-15-0	15%	Yes	0
	SRC-15-90	15%	None	90
	SRCU-15-90	15%	Yes	90

^a S denotes FRP strengthened beam; U denotes inclined U-jacket; first 0, 7.5 and 15 mean prestress level; last 0 and 90 mean corrosion days.

^b The prestressing values were controlled by the force sensor shown in Fig. 5.

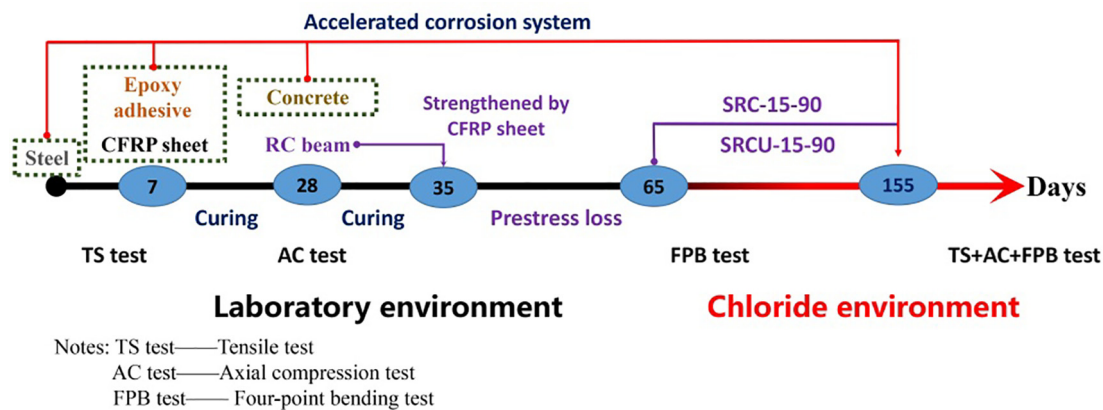


Fig. 4. Timetable of the durability tests.

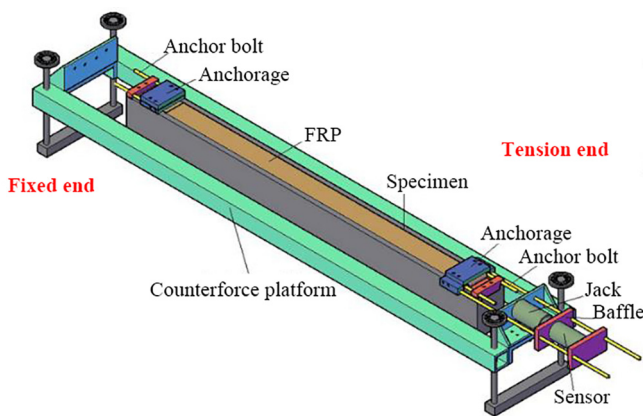


Fig. 5. Prestressing device.

wet-dry cycles (90 days), the tensile strength and elastic modulus of the CFRP reduced by 5.7% and 2.2%, respectively. This degradation can be caused by hydrolysis of the matrix and salt precipitation during the wet-dry cycle weakens the interfacial adhesion of carbon fiber and resin matrix [17,29]. The results indicated that CFRP has a good chlo-

ride ion resistance, which is in accordance with the results of previous studies [4,30]. Different from a single corrosive environment, sustained loads/prestress may lead to an improvement of CFRP properties in a short exposure at elevated temperatures, and the temperatures is benefit to post-curing under the effects of stress redistribution of CFRP [31]. Compared with CFRP, the wet-dry cycling had more negative effects on the tensile strength of the adhesive. As seen in Fig. 9, the tensile strength of the adhesive decreased by approximately 8.3% after a 90-day exposure. This could be attributed to that apart from the penetration of water molecules leads to hydrolysis of the epoxy resin [29], NaCl concentration also has a visible influence on the tensile strength due to soluble compounds in epoxy resin are more easily extracted [17,30].

As regarding to the bare steel bar, Fig. 9 indicates that an exposure of 90 days under the chloride environment reduced the yield strength and ultimate strength by 4.1% and 6.7%, respectively. However, the wet-dry cycling had almost no effect on the elastic modulus of the steel bar. It should be mentioned that the environmental resistance of bare steel bar is different from the steel reinforced in RC beams due to the protection of concrete cover. Thus, reduction coefficients should be considered when using the results of bare steel bar to predict the capacity of RC beams. Additionally, because of hydration of the cement in the humid environment, the compressive strength of the concrete increased from 35 MPa to 38 MPa, as shown in Fig. 9.

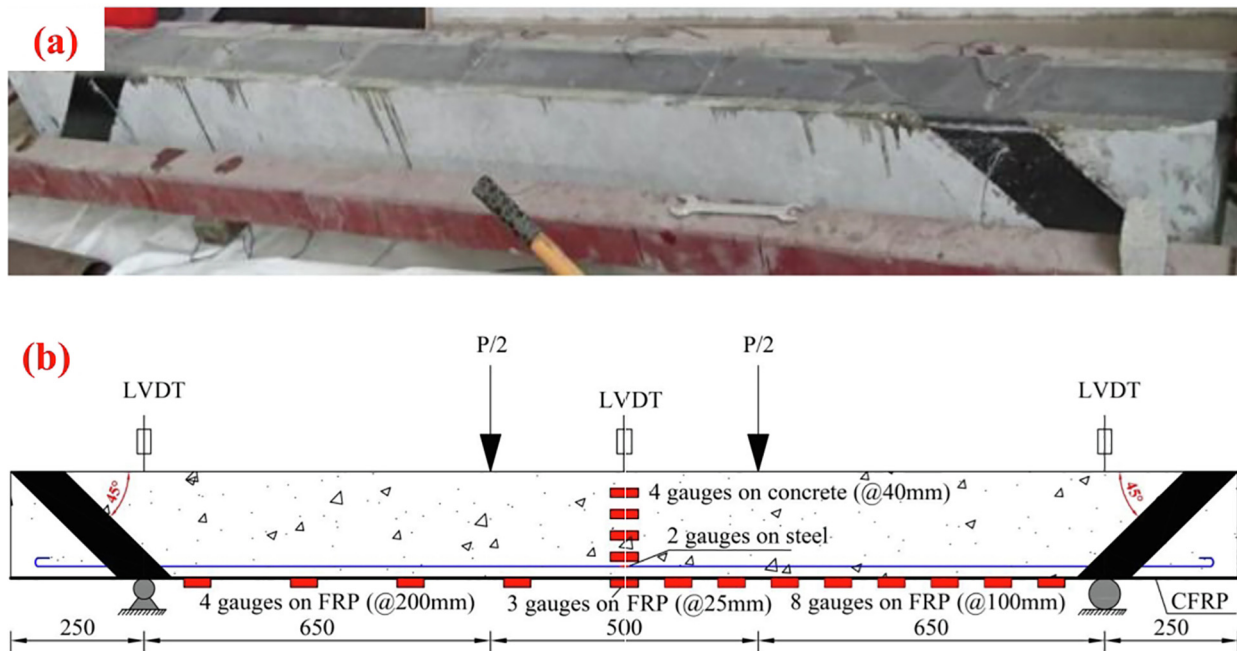


Fig. 6. Instrumentation adopted in the tests with the beams (Units: mm).

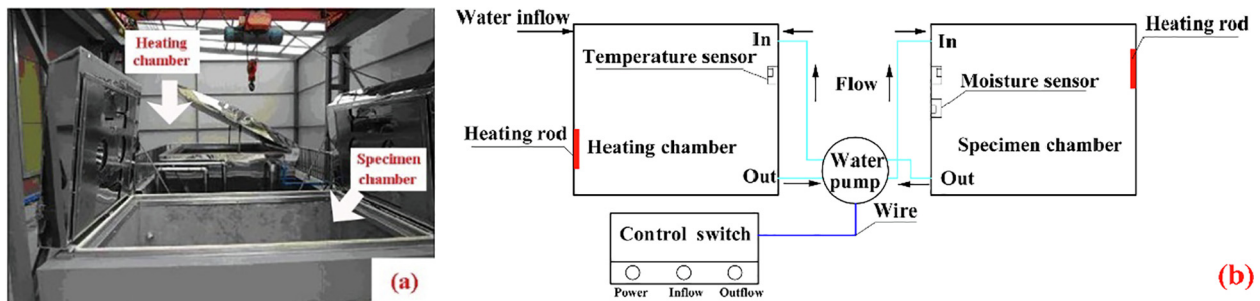


Fig. 7. Corrosion system: (a) physical and (b) sketch diagram.

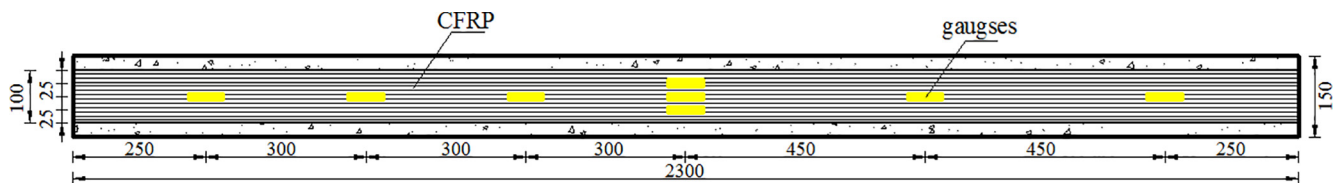


Fig. 8. Position of the strain gauges on CFRP for prestress losses monitor (Unit: mm).

3.2. The time-dependent prestress losses of CFRP

It is well accepted that the prestress losses of RC beam externally strengthened with prestressed CFRP consist of two stages, i.e., anchorage stage and post-releasing stage [32]. Fig. 10 shows the time-dependent characteristics of CFRP prestress losses in the strengthened RC beam. And, the effective prestresses of CFRP in the strengthened beams at a given moment are listed in Table 2. It is noted that the CFRP prestress in Fig. 10 and Table 2 was the average value of the three gauges (Fig. 8) on the mid-span of the CFRP in the strengthened RC beam. It can be found that the prestress loss of CFRP in anchorage stage was not significant (around 2%–8%), which was mainly due to the slip of mechanical anchorage. This result compares well with that reported by the previous studies [7,32]. However, the prestress loss of

CFRP had a rapid increase at the moment of the release of prestressing force, as shown in Fig. 10. In addition, the time-dependent loss cannot be overlooked in post-releasing stage. After 30 days from releasing, the prestress loss was approximately 15% and 12% for SRC-7.5-0 and SRCU-7.5-0, respectively. This loss could be mainly attributed to the coupling effects of concrete shrinkage and the adhesive's post-curing deformation. The results also indicate the advantage of the inclined U-jacket for preventing the prestress loss of CFRP, which can be confirmed by comparing specimens SRC-15-90 and SRCU-15-90, especially within the first 40 days. It is worth mentioning that the prestress loss of SRCU-15-0 shows a sharply increase at the moment of the stress release, which may be deduced by the adhesive resin not fully curing. In general, the prestress losses of CFRP in the strengthened RC beams gradually stabilized after 30 days in post-

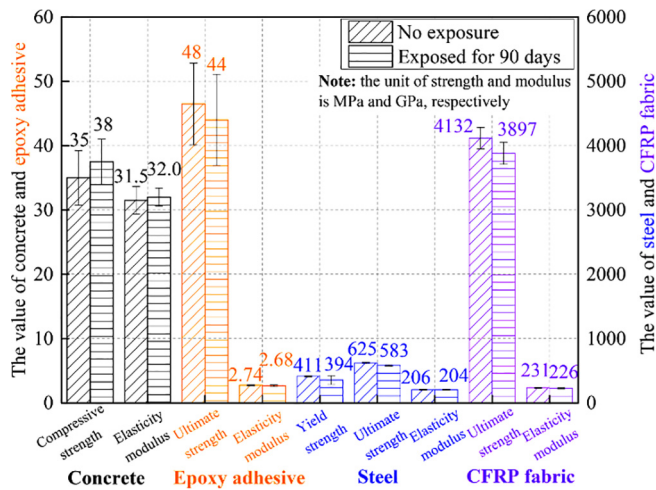


Fig. 9. Mechanical properties of the constituent materials after exposure to wet-dry cycling.

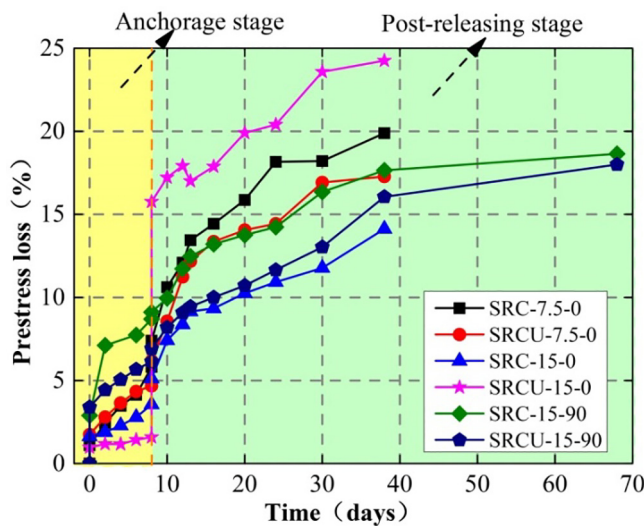


Fig. 10. The relationship between prestress loss and time.

releasing stage, and the value ranged from of 10% to 15%, as shown in Fig. 10.

Fig. 10 also shows the effect of exposure time in the chloride-containing environment on the prestress losses of the CFRP. A comparison of the specimens in Fig. 10 demonstrates that the prestress loss of SRC-15-90 was higher than SRC-15-0, indicating that wet-dry cycling environment is adverse to the prestress, but the negative influence was not significant. This may be mainly due to the hydrolysis reaction of adhesive resin in such chloride environment, as aforementioned in Section 3.1, resulting in a further post-curing deformation of the adhe-

sive. After a 60-day exposure in post-releasing stage, the prestress loss of the CFRP-strengthened RC beams strengthened were approximately 18%.

3.3. Cracking and loading capacity

The cracking, yielding, and ultimate loads of the specimens under four-point bending are shown in Fig. 11. As presented in Fig. 11, the cracking load of the RC beam after strengthened with 15% prestressed load (SRC-15-0) increased from 4.6 kN-m to 7.1 kN-m. However, as the specimens with 0 (SRC-0-0) and 7.5% (SRC-7.5-0) prestressed load, the cracking load almost unchanged compared with the control RC beam. The results may be explained that CFRP with low prestressing level provides limited constraints for the flexural performance of the RC beams, i.e., low utilisation efficiency of CFRP [7,9]. A different phenomena can be found for the strengthened specimens with inclined U-jacket named SRCU-0-0 and SRCU-7.5-0, the cracking load gradually increased to 5.0 kN-m and 5.8 kN-m. Interestingly, the cracking load of the specimen SRCU-15-0 increased to 6.0 kN-m, which was lower than that of the specimen SRC-15-0 (7.1 kN-m). The lower cracking load of SRCU-15-0 can be attributed to the performance variation of the concrete [3].

As regarding to the yield load, as shown in Fig. 11, compared with the control RC beam (12.0 kN-m), the specimens strengthened with CFRP sheet increased to 16.5, 16.8 and 17.0 kN-m for specimens SRC-0-0, SRC-7.5-0 and SRC-15-0, respectively. This result indicates that CFRP sheet enhanced the flexural performance of the RC beam. For the specimens with inclined U-jacket, the yield loads increased from 12.0 kN-m to 14.5, 16.0, and 17.8 kN-m for the specimens with 0% (SRCU-0-0), 7.5% (SRCU-7.5-0) and 15% (SRCU-15-0) prestressed load, respectively. It can be concluded from these findings that the prestressing of CFRP can enhance the yield load of RC beam, and this advantage could be further improved by the combination of high-level prestressing and inclined U-jacket.

In addition to the cracking load and yield load, the ultimate load increased rapidly with an increase in the prestressed load. As shown in Fig. 11, the ultimate load increased from 16.0 kN-m (control RC beam) to 22.5 kN-m and 25.7 kN-m for the RC beam strengthened with CFRP sheet with 0% (SRC-0-0) and 7.5% (SRC-7.5-0) prestressed load, respectively. Similar trend can be observed in specimens SRC-0-90 and SRC-15-90.

Fig. 11 also demonstrates the influence of wet-dry cycling on the CFRP strengthening system. For the specimens exposed to the wet-dry cyclic condition after 90 days, it can be found that the cracking load of specimen SRC-0-90 (4.5 kN-m) increased by 9.8% compared to SRC-0-0 (4.1 kN-m), and the cracking load of specimen SRC-15-90 was close to that of SRC-15-0. Similarly, the yield loads of SRC-15-90 and SRCU-15-90 were almost unchanged compared with SRC-15-0, and SRCU-15-0, respectively. In addition, the ultimate load of SRC-0-90 was unchanged compared to SRC-0-0, and even the value of SRC-15-90 increased by 6.1% compared with SRC-15-0. These findings indicate that the wet-dry cycling did not cause a significant adverse effect on the flexural performance of the strengthened RC beams, which may be due to the short exposure period in the test. As for the experimental phenomenon of the improvement of the strengthened RC beams after

Table 2

Effective prestresses of CFRP in the strengthened beams.

Specimen	Prestressing force (kN)	Initial prestress (MPa)	Duration in post-releasing stage (Days)	Effective Prestress (MPa)
SRC-7.5-0	5.2	563.2	30 (Room temperature)	451.4
SRCU-7.5-0	5.2	641.5	30 (Room temperature)	530.7
SRC-15-0	10.4	944.2	30 (Room temperature)	810.8
SRCU-15-0	10.4	860.8	30 (Room temperature)	652.0
SRC-15-90	10.4	762.6	60 (Chloride-containing environment)	620.4
SRCU-15-90	10.4	769.9	60 (Chloride-containing environment)	635.1

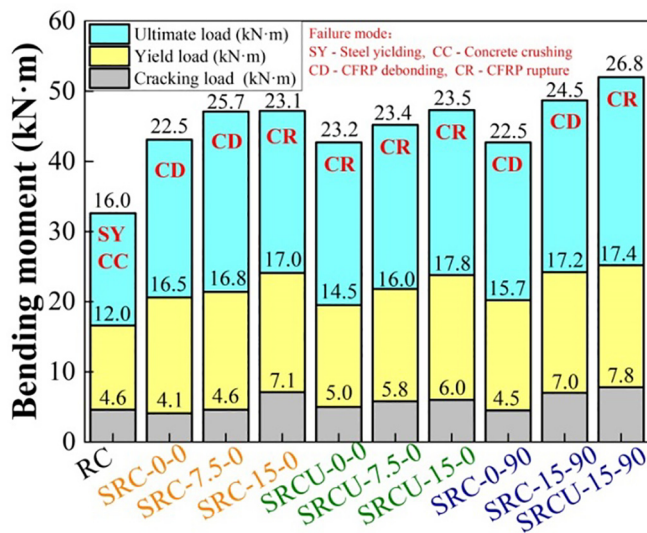


Fig. 11. Experimental results of loading capacity and failure modes.

exposure, it could be attributed to that apart from the slight reduction in the mechanical properties of the CFRP and the steel bar after a 90-day exposure, the post-exposure enhancement of the concrete in the strengthened RC beam may be the main reason [31], as shown in Fig. 9. Additionally, it can be found from Fig. 11 that the ultimate load of SRCU-15-90 increased by 9.4% compared with SRC-15-90. The reason for this increase is not only because of the higher effective prestress level of the former (Table 2), but also the strengthening effect of the inclined U-jacket.

3.4. Failure mode

The RC beams strengthened with CFRP sheet exhibited similar failure processes during the bending test. Vertical micro-cracks appeared firstly in the pure bending zone at the bottom of the beam. The crack widths increased and developed gradually at a certain interval towards the loading point with the increased load. Cracks later appeared in the bending-shear area at the bottom of the beam. A diagonal crack appeared near the bottom of the bending and bending-shear cracks, and the trend to the horizontal was more obvious nearer the CFRP sheet, resulting into the local debonding of the CFRP. Similar test results can also be found in literature [7,33]. As the load increased, these horizontal cracks gradually propagated to the support and then the steel yielded, and finally a “BANG” sound was heard due to CFRP fracture or CFRP debonding when the bearing capacity reached. The failure modes of specimens are shown in Fig. 12.

As illustrated in Fig. 12a, the control RC beam exhibited balanced-reinforced failure mode types: steel yielding followed by concrete crushing. Compared with the control RC beam, the prestressed-CFRP strengthened RC beam had more cracks, and the cracks width were smaller. This finding indicates that the prestressed CFRP led to the strengthened beam more evenly stressed under bending load, and consequently enhanced the flexural performance of the RC beam, which can be confirmed by the cracking load discussed in Section 3.3. It can be found from Fig. 12 that the specimens with 0% (SRC-0-0) and 7.5% (SRC-7.5-0) without prestressed load failed by CFRP debonding, while the specimen with 15% prestressed (SRC-15-0) failed by CFRP rupture. These findings suggest that as there is no end anchorage, the CFRP strengthened beams with/without prestressed load are prone to interface peeling failure; however, CFRP rupture failure could occur as the prestressed load increased. This could be attributed to the fact that it is easy to reach the ultimate

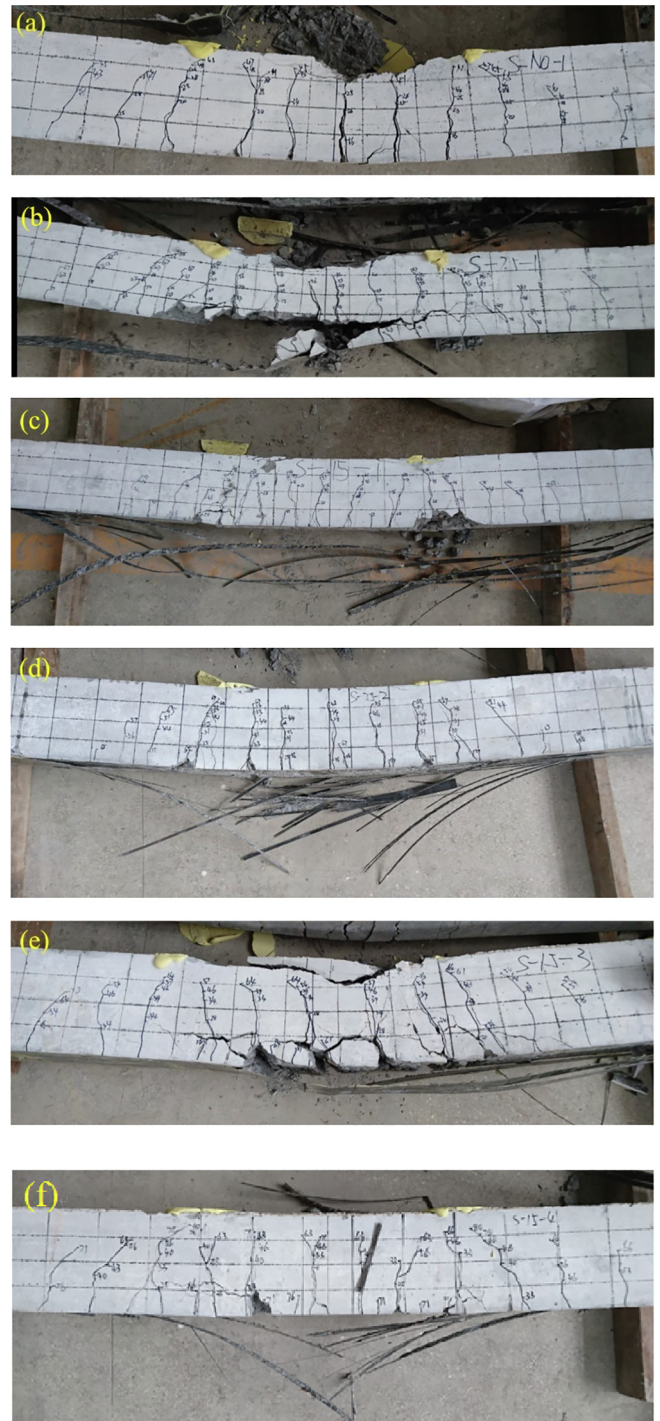


Fig. 12. Failure modes of the beams: (a) control RC beam; (b) SRC-7.5-0; (c) SRC-15-0; (d) SRCU-7.5-0; (e) SRC-15-90 and (f) SRCU-15-90.

strength of CFRP under the combined action of high-level prestress and external loads.

Interestingly, it can be observed from Fig. 12 that inclined U-jackets could have a significant influence on the failure mode of the prestressed-CFRP strengthened beams. A comparison of SRC-15-0, SRCU-0-0, SRC-7.5-0 and SRCU-7.5-0 shows that the addition of inclined U-jackets changed the failure modes of the strengthened beam from debonding failure to CFRP rupture. This is because that the end constraint induced by inclined U-jackets can efficiently prevent the local debonding near the anchorage, leading to the CFRP rupture prior

to the anchorage failure, and consequently reduced the possibility of the entire CFRP sheet peeling failure [34]. Similar result can be found by comparing SRC-15-90 and SRCU-15-90.

The influence of the exposure environment on the failure modes of the strengthened beams with prestressed CFRP could be found in Fig. 12. A comparison of SRC-15-0 and SRCU-15-90 suggests that the failure mode of the strengthened beams with prestressed CFRP may be changed by the exposure environment, i.e., from CFRP rupture to interface debonding. This change in failure mode could be attributed to that water penetration led to hydrolysis of the adhesive layer, which reduced the chemical adhesive force and mechanical force between CFRP sheet and concrete substrate [35]. Thus, the CFRP peeling of the specimen SRC-15-90 was prior to the CFRP rupture. However, as the end constraints of the inclined U-jackets, specimen SRC-15-90 failed by CFRP rupture rather than debonding, as shown in Fig. 12f.

3.5. Load-deflection curves

The load-deflection curves of the RC beams are illustrated in Fig. 13. As can be seen in Fig. 13, the load-deflection curves for specimens with different prestressed load levels. Prior to the RC beam cracking, the load-deflection curve trend of the RC beam strengthened with prestressed CFRP sheet is seen to be the same as that of the ordinary CFRP sheet reinforced specimen. However, the beam shows better deformation characteristics than the CFRP sheet strengthened beam without a prestressed load. It is evident that prestressed loading improved the reinforcement and performance of the RC beam.

The effects of the inclined U-jacket on load-deflection are illustrated in Fig. 13c. The inclined U-jacket effectively constrained CFRP sheet, and the deformation gradually decreased as the prestress load levels increased. This result can be attributed to the increased effective prestress of CFRP sheet.

The performances of specimens exposed to the corrosion environment after 90 days are shown in Fig. 13d, the flexural stiffness of the SRC-0-90 after cracking decreased compared with the specimen SRC-0-0, which indicates the degradation of the bonding behaviour occurred. However, this result is not fit for the specimens with prestressed load. Compared with SRC-15-0, the flexural stiffness of SRC-15-90 after cracking did not reduce, and even increased after steel yielding. This may be because that the coupling effect of environment exposure and prestressed load could improve the performance of CFRP in a given time [31]. This finding also indicates the suitability of prestressed CFRP in the strengthening of RC beams exposed to chloride environments.

3.6. Strain variations of concrete

The strain distribution of the beam under various loads is illustrated in Fig. 14, and it depicts that the strain is linear in the pure bending section of specimens prior to concrete cracking, and that the strain distribution along the height of the beam section basically meets the assumption of the plane section. However, the concrete in the tension zone failed to work after cracking, and the strain gauges record failure, although the strain at other points maintained the assumption of the plane section. There was an increased strain growth

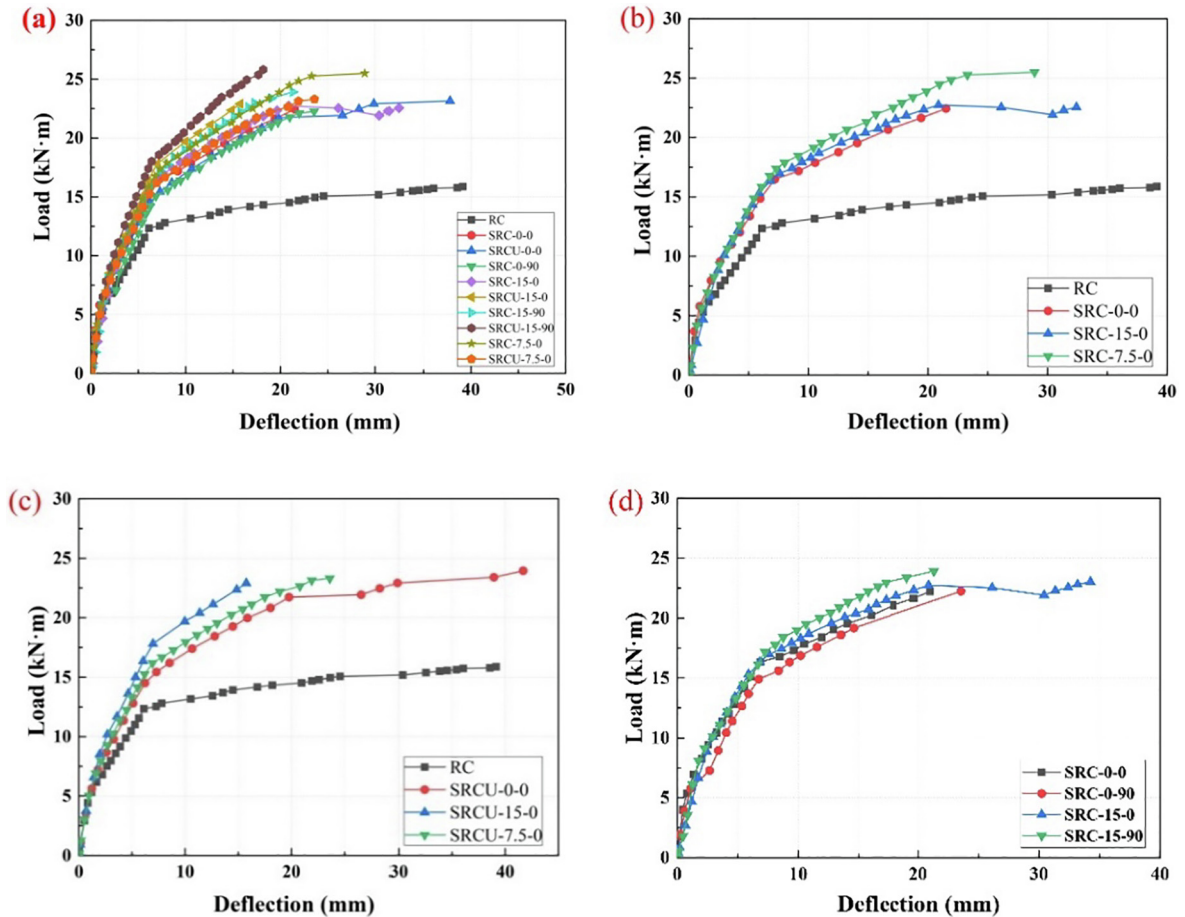


Fig. 13. Load-deflection curves, (a) all test pieces; (b) without corrosion; (c) specimens with U-jacket; (d) exposed for 90 days.

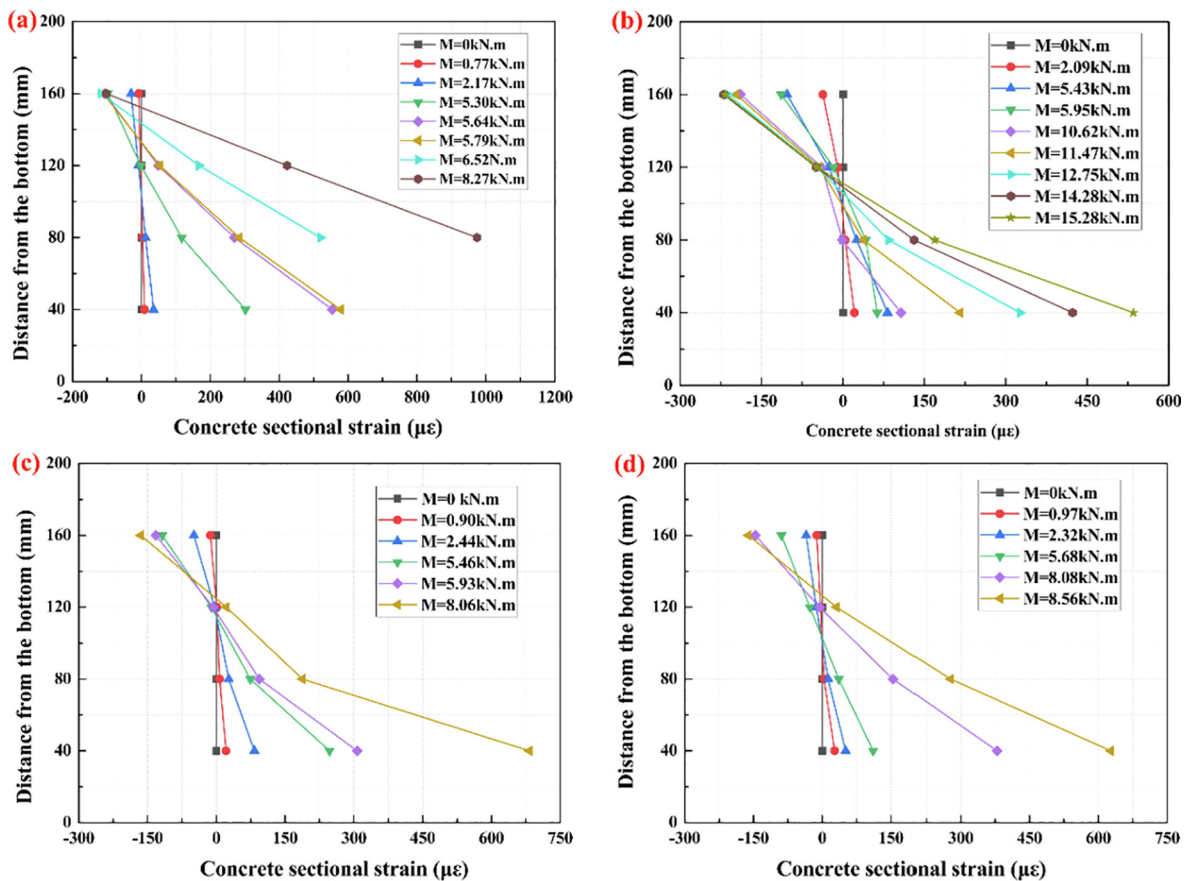


Fig. 14. Strain changes of concrete, (a) RC; (b) SRCU-0-0; (c) SRCU-15-0 and (d) SRCU-15-90.

rate of the strain gauges at the 80 mm and 120 mm locations. Compared with specimens SRCU-0-0 (Fig. 14b) and SRCU-15-0 (Fig. 14c), the strain of the concrete decreased with the same load level, which indicates the effects of prestressed loading. As the prestressing load level increased, the neutral axis of the beam rose, as shown in Fig. 14b and c. Fig. 14d depicts the strain of the concrete with the effect of immersion. It can be found that SRCU-15-90 shows a lower strain compared to SRCU-15-0 at approximately 5.5 kN.m, which is probably related to the increased properties of the concrete.

3.7. Strain variation of CFRP

Fig. 15 illustrates the relationship between applied loading and the strain of CFRP sheet, where the abscissa is the distance from the strain gauge to the centre of the beam. As depicted in Fig. 15a, the strain of CFRP sheet increases slowly and is essentially zero in the initial stage. CFRP sheet begins to bear the loading after concrete cracking, the largest increase in strain occurs in the mid-span, and there is a gradual decrease towards both ends. After the steel yields, the strain begins to increase sharply, and CFRP sheet peeling-off or rupture occurred with respect to the failure with loading.

A comparison of SRC-0-0 and SRCU-15-0 illustrated in Fig. 15a and b, respectively, showing that the strain of SRCU-15-0 is relatively lower than that of SRC-0-0 at the same loads. This is mainly due to the decompression effect of prestressed load on the strengthened beams. However, the actual strain of the CFRP in SRCU-15-0 was equal to the value caused by the combined action of the prestressed load and bending load, which might be higher than the strain of SRC-0-0 caused by the sole effect of bending load. Thus, the CFRP sheet of SRCU-15-0 ruptured as the strain reached its limited value.

The influence of environment exposure on the distribution of CFRP strain can be found by comparing SRC-0-0 and SRC-0-90, SRCU-15-0 and SRCU-15-90, as shown in Fig. 15. Interestingly, with the increasing of bending load, the specimen after exposure had a more uniform distribution than that without exposure. This result suggests that this exposure led to a faster transfer of the bond stress between CFRP and concrete from mid-span to CFRP end. It could be attributed to the degradation of interface performance after exposure [27].

The strain and utilisation efficiency of CFRP sheet at concrete cracking, steel yield and in the ultimate stages are listed in Table 3. In this study, the utilisation efficiency of CFRP is defined as the ratio of the CFRP strain to its ultimate strain. It can be seen from Table 3 that the utilisation efficiency increases with an increase in the applied load, and the utilisation ratios of specimen SRC-0-0 for the cracking, yielding, and ultimate stages are 3.3%, 18.7% and 56.6%, respectively. However, for specimen with a prestressed load of 7.5%, the utilisation efficiency of specimen SRC-7.5-0 is 86.2% and the ratios at concrete cracking and steel yield stages are 17.4% and 44.9%. The ratio at the ultimate stage is close to the ultimate tensile strain of the CFRP sheet. These results indicate that a prestressed load is beneficial to the use of CFRP sheet [9,13].

With respect to the effect of the inclined U-jacket, the utilisation efficiencies of SRCU-0-0 at cracking, yielding and in the ultimate state were 9.4%, 28.1% and 84.7%, respectively, which is increased by 184.9%, 50.3% and 49.6% compared with SRC-0-0. The increased utilisation efficiency of CFRP sheet is attributed to the inclined U-jacket at the end of the bonding interface [3]. The utilisation efficiency of SRCU-0-0 in the ultimate stage is close to that of SRC-7.5-0, which indicates that the inclined U-jacket plays a similar role to that of a prestressing load in improving the utilisation efficiency of the CFRP sheet.

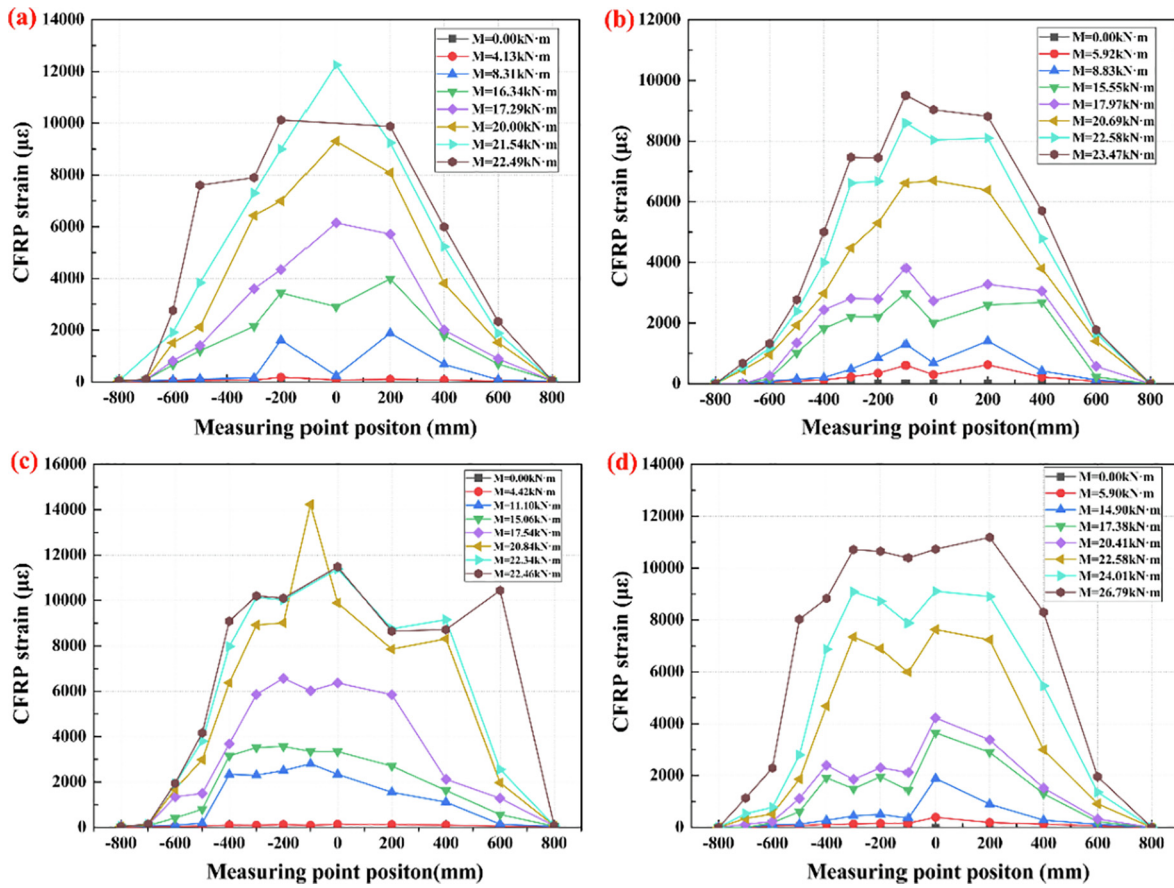


Fig. 15. Strain variations of CFRP: (a) SRC-0-0; (b) SRCU-15-0; (c) SRC-0-90 and (d) SRCU-15-90.

Table 3
Strain and utilisation efficiency of CFRP sheet at different stages.

Code	Initial stage		Cracking stage		Yield stage		Ultimate stage	
	Strain ($\mu\epsilon$)	Utilisation efficiency (%)	Strain ($\mu\epsilon$)	Utilisation efficiency (%)	Strain ($\mu\epsilon$)	Utilisation efficiency (%)	Strain ($\mu\epsilon$)	Utilisation efficiency (%)
SRC-0-0	0	0	571	3.3	3347	18.7	10,125	56.6
SRC-7.5-0	2600	14.5	2600 + 516	17.4	2600 + 5425	44.9	2600 + 12828	86.2
SRCU-0-0	0	0	1696	9.4	5027	28.1	15,160	84.7
SRC-0-90	0	0	352	2.0	3633	21.0	10,206	59.0

It can be also seen from Table 3 that the utilisation efficiency of the specimen exposed to the chloride ion environment (SRC-0-90) was about 3% different from that of the specimen SRC-0-0. Moreover, there was a slight increase in the utilisation efficiency at the yield and ultimate stages. These findings indicate that short-term chloride ion exposure has no obvious effect on the utilisation ratio of CFRP sheet in strengthening RC beams.

4. Calculation for load-bearing capacity

According to different design parameters and constituent materials of the strengthening system, it is not surprising that the prestressed-CFRP strengthened RC beams might have different failure modes [8], including: 1) Steel bars yield followed by CFRP rupture and concrete crush; 2) CFRP-concrete interface debonding before CFRP rupture; 3) CFRP rupture and concrete crush, but steel bar did not yield; 4) Steel bars yield and concrete crush, but CFRP did not rupture; 5) CFRP rupture, but steel bar did not yield and concrete did not crush; 6) Concrete

cover fall off followed by steel bar yield. In general, failure mode 1 and mode 2 are the main failure modes of prestressed-CFRP strengthened RC beams [7,32,36]. However, CFRP debonding failure is characterized by brittle failure, which could be avoided by fixing anchors in engineering [8]. Thus, this study mainly analyzes the flexural theory of prestressed-CFRP strengthened RC beams failed by mode 1.

4.1. Fundamental assumption

In accordance to the previous studies [37] on the calculation of CFRP-strengthened RC beam based on classical beam theory, the following assumptions are used in this study:

- (1) The deformation of the concrete section of the prestressed-CFRP strengthened RC beam conforms to the assumption of the plane section, which is confirm by the above discussion.
- (2) Assumption that the bonding between CFRP and concrete is good enough, and without relative slip. This is because that the failure mode of CFRP rupture is focused in this study.

- (3) Without regard to the tensile strength of the concrete, and the action of compression steel bars after the crack at the bottom of the beam.

In terms of the constitutive relation of steel, an ideal elastoplastic model for the stress–strain relationship is adopted. In addition, linear elastic model is used for the CFRP sheet. According to the Chinese guideline ‘Code of design of concrete structures’ [38], the compressive stress–strain relationship of concrete described by Eqs. (1)–(5) is used in this study.

$$\text{As } \varepsilon_c \leq \varepsilon_0,$$

$$\sigma_c = f_c \left[1 - \left(1 - \frac{\varepsilon_c}{\varepsilon_0} \right)^n \right] \quad (1)$$

$$\text{As } \varepsilon_0 < \varepsilon_c \leq \varepsilon_{cu},$$

$$\sigma_c = f_c \quad (2)$$

$$n = 2 - \frac{1}{60} (f_{cu,k} - 50) \quad (3)$$

$$\varepsilon_0 = 0.002 + 0.5 (f_{cu,k} - 50) \times 10^{-5} \quad (4)$$

$$\varepsilon_{cu} = 0.0033 - 0.5 (f_{cu,k} - 50) \times 10^{-5} \quad (5)$$

where σ_c is the compressive stress of concrete as the strain is ε_c , f_c is design value of axial compressive strength of concrete, ε_0 is the strain of concrete as the compressive stress of concrete up to f_c , the value takes to 0.002 as the calculated value of ε_0 less than 0.002, n is the coefficient, the value takes to 2.0 as the calculated value of n less than 2.0, ε_{cu} is the ultimate compressive strain of concrete, the value takes to 0.0033 as the calculated value of ε_{cu} greater than 0.0033.

4.2. Stress analyzes of the strengthened RC beam during loading

- (1) *Prestressing stage:* The effective prestress obtained by CFRP can be expressed as:

$$\sigma_{cf,0} = \sigma_{pe} = \sigma_{con} - \sigma_{l1} - \sigma_{l2} - \sigma_{l3} - \sigma_{l4} \quad (6)$$

where σ_{con} is the design stress level, σ_{l1} is the loss of prestress caused by technology and anchorage slip, σ_{l2} is the loss of prestress caused by shrinkage deformation of concrete at moment of release, σ_{l3} is the loss of prestress caused by relaxation of CFRP sheet, and σ_{l4} is the loss of prestress caused by exposure environment.

Epoxy resin was used at the bottom of the beam, and the forced was released after the epoxy resin post cured. The loss of prestress at this moment led to eccentric compressive stress, resulting in the beam generate arch value. The strain and stress of the concrete cross section is shown in Fig. 16, and the preloading stress can be calculated by the following formula:

$$\sigma_{c,0} = \frac{N_{pe}}{A_0} + \frac{N_{pe}e_0y_c}{I_0} \quad (7)$$

where N_{pe} is Effective pretension of CFRP, $N_{pe} = \sigma_{pe} \times A_{cf}$, σ_{pe} can be obtained by Eq. (6). e_0 is the distance between center of resultant force of CFRP sheet and the neutral axis of the beam, y_c is the distance of the bottom of the beam to neutral axis of the beam. A_0 is the conversion

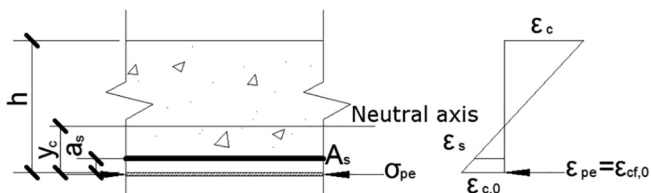


Fig. 16. Stress and strain of the beam after release loading.

area of the beam, can be obtained by Eq. (8), and I_0 is the moment of inertia of concrete beam, can be calculated by Eq. (9).

$$A_0 = \frac{(bh - A_s)E_c + A_s\psi_sE_s}{E_c} = bh + \alpha_{Es}A_s \quad (8)$$

where E_s and E_c are the tensile modulus of steel and concrete, A_s is the area of the steel section, α_{Es} is conversion coefficient, and $\alpha_{Es} = \psi_sE_s/E_c - 1$. ψ_s is the reduction factor for the tensile properties of the steel bar due to environmental exposure. According to the test results as shown in Fig. 11 and previous studies [38,40], ψ_s is approximately 0.95 in marine environment.

$$I_0 = \frac{1}{12}bh^3 + \left(\frac{h}{2} - y_c \right)^2 bh + \alpha_{Es}A_s(y_c - a_s)^2 \quad (9)$$

- (2) *Loading until cracking:* There are three stages in the process of loading to the concrete cracking at the bottom of the beam: decompression state stage, normal use stage and concrete cracking stage.

At decompression state stage, the stress of CFRP sheet equal to the sum of initial effective prestress and compressive stress of concrete, which can be presented as:

$$\sigma_{cf,1} = \sigma_{pe} + \sigma_{c,0} \quad (10)$$

where $\sigma_{cf,1}$ is the stress of CFRP, σ_{pe} is initial effective prestress of CFRP, and $\sigma_{c,0}$ is compressive stress of concrete.

The applied loading makes the beam into the bending state, and the tensile stress increased at the bottom of the beam during loading, when the tensile stress at the beam bottom equal to CFRP prestressed compressive stress, stress of the concrete beam bottom is zero. This process is eliminated pressure stage. The stress and strain states of concrete is shown in Fig. 17.

Based on Fig. 17:

$$\sigma_{c,0} - \frac{M_1y_c}{I_1} = 0 \quad (11)$$

where I_1 is moment of inertia of CFRP and RC beam. The conversion area A_1 and I_1 of the beam and CFRP sheet is:

$$A_1 = A_0 + \alpha_{Ecf}A_{cf} \quad (12)$$

$$\alpha_{Ecf} = \frac{\psi_{cf}E_{cf}}{E_c} - 1 \quad (13)$$

$$y_c = \frac{\frac{1}{2}bh^2 + \alpha_{Es}A_s a_s}{bh + \alpha_{Es}A_s + \alpha_{Ecf}A_{cf}} \quad (14)$$

$$I_1 = I_0 + \alpha_{Ecf}A_{cf}y_c^2 \quad (15)$$

where E_{cf} is the tensile modulus of CFRP, A_{cf} is the cross-section area of CFRP, ψ_{cf} is the reduction coefficient of the CFRP under chloride environment, which could be 0.95 according to the previous studies [29,41]. And then,

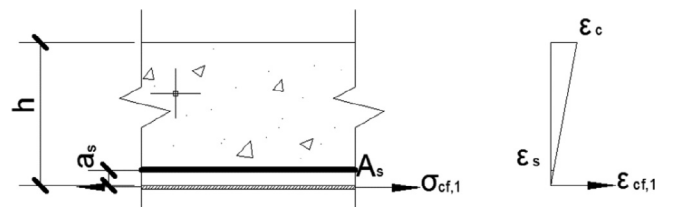


Fig. 17. Stress and strain of the beams at decompression state.

$$M_1 = \frac{\sigma_{c,0} I_1}{y_c} \quad (16)$$

The stress and strain of CFRP can be obtained by:

$$\sigma_{cf,1} = \sigma_{cf,0} + \psi_{cf} E_{cf} \varepsilon_{c,0} = \sigma_{cf,0} + \sigma_{c,0} \quad (17)$$

$$\varepsilon_{cf,1} = \frac{\sigma_{cf,1}}{\psi E_{cf}} = \varepsilon_{cf,0} + \varepsilon_{c,0} \quad (18)$$

Normal use stage: After decompression state, the tensile stress of the concrete at the bottom of the beam increased with loading, cracks appeared as the tensile stress exceeds the tensile strength of the concrete.

Concrete cracking stage: After concrete cracking, the concrete in the tensile zone gradually out of work and its tensile strength rapidly decreases. According to the basic assumption, the tensile strength of concrete can be ignored. As the crack expands, the neutral axis of the beam section gradually rises to the top of the beam. Since the increment of the neutral axis is difficult to calculate and the increment is small, the influence of the increment is ignored, and the position of the neutral axis is assumed to be unchanged after cracking [42]. The bending moment at concrete cracking can be obtained by [38]:

$$M_{cr} = M_1 + \frac{(0.7 + \frac{120}{h}) \gamma_m f_{tk} I_1}{y_c} \quad (19)$$

$$= \frac{I_1}{y_c} [\sigma_{c,0} + (0.7 + \frac{120}{h}) \gamma_m f_{tk}]$$

where M_1 is the decompression state bending moment, γ_m is the value of plastic influence coefficient of the resistance moment of concrete member section, generally equal to 1.55.

(3) After cracking to steel yield stage: The stress of CFRP can be expressed as:

$$\sigma_{cf,2} = \sigma_{cf,1} + \sigma_{cf,y} = \sigma_{pe} + \sigma_{c,0} + \sigma_{cf,y} \quad (20)$$

where $\sigma_{cf,1}$ is the stress of CFRP at decompression state stage, $\sigma_{cf,y}$ is the stress increment of CFRP from decompression state stage to steel yield stage. The equivalent stress is shown in Fig. 18, and two parameters of α_1 and β_1 is introduced.

$$C = \alpha_1 f_c b x = \alpha_1 \kappa f_c b \beta_1 x_0 \quad (21)$$

$$\beta_1 = \frac{x}{x_c} \quad (22)$$

$$h_c = x_c - \frac{x}{2} \quad (23)$$

where α_1 is the ratio of the stress of concrete in compression zone to axial compressive strength of concrete (f_c), β_1 is the ratio of equivalent compression zone height (x) to depth of compression zone (x_c), h_c is distance from the net force to the neutral axis. κ is increase coefficient of concrete strength. Existing research results confirmed that after to 200 wet-dry cycles in various types of salt solutions, the concrete was no damage, and even, the performance of the concrete was better [18,43]. Based on the database in references [17,45,45], κ equals to

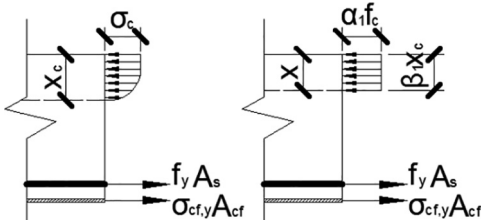


Fig. 18. Equivalent stress.

1.1 in a chloride environment. And, C , h_c , α_1 and β_1 can be obtained by Chinese standard (GB50010-2002) [38]:

As $0 \leq \varepsilon_c \leq \varepsilon_0$:

$$C = \left[\frac{\varepsilon_c}{\varepsilon_0} - \frac{1}{3} \left(\frac{\varepsilon_c}{\varepsilon_0} \right)^2 \right] \kappa f_c b x_c \quad (24)$$

$$h_c = \frac{8\varepsilon_0 - 3\varepsilon_c}{12\varepsilon_0 - 4\varepsilon_c} x_c \quad (25)$$

$$\alpha_1 = \frac{2}{3} \cdot \frac{\frac{\varepsilon_c}{\varepsilon_0} \left(3 - \frac{\varepsilon_c}{\varepsilon_0} \right)^2}{4 - \frac{\varepsilon_c}{\varepsilon_0}} \quad (26)$$

$$\beta_1 = \frac{4\varepsilon_0 - \varepsilon_c}{6\varepsilon_0 - 2\varepsilon_c} \quad (27)$$

As $\varepsilon_0 \leq \varepsilon_c \leq \varepsilon_{cu}$:

$$C = \left(1 - \frac{\varepsilon_0}{3\varepsilon_c} \right) \kappa f_c b x_c \quad (28)$$

$$h_c = \frac{\frac{1}{2} - \frac{1}{12} \left(\frac{\varepsilon_0}{\varepsilon_c} \right)^2}{1 - \frac{1}{3} \cdot \frac{\varepsilon_0}{\varepsilon_c}} x_c \quad (29)$$

$$\alpha_1 = \frac{2}{3} \cdot \frac{\left(3 - \frac{\varepsilon_0}{\varepsilon_c} \right)^2}{6 - 4 \frac{\varepsilon_0}{\varepsilon_c} + \left(\frac{\varepsilon_0}{\varepsilon_c} \right)^2} \quad (30)$$

$$\beta_1 = \frac{1 - \frac{2}{3} \cdot \frac{\varepsilon_0}{\varepsilon_c} + \frac{1}{6} \left(\frac{\varepsilon_0}{\varepsilon_c} \right)^2}{1 - \frac{1}{3} \cdot \frac{\varepsilon_0}{\varepsilon_c}} \quad (31)$$

After the concrete cracks, the reinforcing bars of the beam work together with CFRP until the steel bar yields. The stress and strain states of concrete beams at the yield of reinforcement are shown in Fig. 19.

$\sigma_s = f_y$ can be used as steel bars yield, the strain of steel is ε_y , based on Fig. 19,

$$\frac{\varepsilon_y}{h_0 - x_c} = \frac{\varepsilon_{cf,y}}{h - x_c} = \frac{\varepsilon_c}{x_c} \quad (32)$$

where $\varepsilon_{cf,y}$ is the strain increment of CFRP from decompression state stage to steel yield stage, and the strain of CFRP at steel bars yield is:

$$\varepsilon_{cf,2} = \varepsilon_{cf,1} + \varepsilon_{cf,y} = \varepsilon_{cf,0} + \varepsilon_{c,0} + \varepsilon_{cf,y} \quad (33)$$

Based on Eqs. (24) and (28), C_1 and C_2 can be calculated:

$$C_1 = \psi_s f_y A_s + \psi_{cf} E_{cf} \varepsilon_{cf,2} A_{cf} \quad (34)$$

$$C_2 = \begin{cases} \left[\frac{\varepsilon_c}{\varepsilon_0} - \frac{1}{3} \left(\frac{\varepsilon_c}{\varepsilon_0} \right)^2 \right] \kappa f_c b x_c, & 0 \leq \varepsilon_c \leq \varepsilon_0 \\ \left(1 - \frac{\varepsilon_0}{3\varepsilon_c} \right) \kappa f_c b x_c, & \varepsilon_0 < \varepsilon_c \leq \varepsilon_{cu} \end{cases} \quad (35)$$

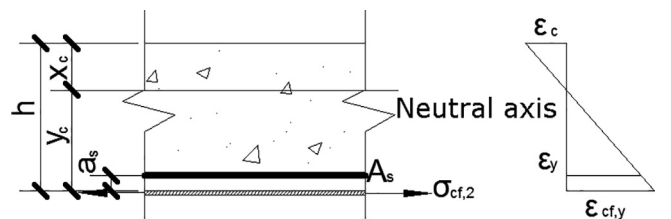


Fig. 19. Stress and strain of beams when steel bars yield.

According to the balance of forces, let $C_1 = C_2$, based on Eqs. (32)–(35), ε_c can be obtained, and substitute into (26), (27) and (30), (31), α_1 and β_1 can be obtained. Based on Eq. (22), x can be obtained. Then, yield load of RC beams strengthened with prestressed CFRP can be obtained according to the balance relationship of moment of section bending, as follows:

$$M_y = \psi f_y A_s \left(h_0 - \frac{x}{2} \right) + \psi_{cf} E_{cf} \varepsilon_{cf,2} A_{cf} \left(h - \frac{x}{2} \right) \quad (36)$$

(4) Steel yield until failure stage: CFRP bear the load after steel yield, and the increases of the strain increased, CFRP and concrete interface shear stress increased. The development of the interface crack speed is accelerated, the cracks in the main span and support extended, and leads to the local deboning of CFRP. As the load continues to increase, the expansion of interfacial cracks will lead to CFRP debonding failure of the specimens without anchorage, but CFRP ruptured as the specimens with end anchorage. The stress of CFRP can be expressed as:

$$\sigma_{cf,3} = \sigma_{cf,1} + \sigma_{cf,u} = \sigma_{pe} + \sigma_{c,0} + \sigma_{cf,u} \quad (37)$$

where $\sigma_{cf,1}$ is the stress of CFRP at decompression state stage, $\sigma_{cf,u}$ is the stress increasement of CFRP from decompression state stage to failure. The stress and strain states of concrete beams are shown in Fig. 20.

Based on Fig. 20,

$$\frac{\varepsilon_{cu}}{x_b} = \frac{\varepsilon_{cf,u}}{h - x_b} \quad (38)$$

$$\xi_b = \frac{x_b}{h} = \frac{\varepsilon_{cf,u}}{\varepsilon_{cu} + \varepsilon_{cf,u}} \quad (39)$$

$$x = \beta_1 x_b \quad (40)$$

$$\alpha_1 k_f c b x = \varepsilon_s A_s + \psi_{cf} E_{cf} A_{cf} \varepsilon_{cf,3} \quad (41)$$

where x_b is limit pressure zone height, ξ_b is relative limit compression zone height.

As $\xi < \xi_b$, the failure mode is CFRP rupture, but the ultimate compressive strain of concrete in the compression zone is not reached, then,

$$\alpha_1 E_c \varepsilon_c b x = \psi f_y A_s + \psi_{cf} E_{cf} A_{cf} \varepsilon_{cf,3} \quad (42)$$

$$M_u = \psi f_y A_s \left(h_0 - \frac{x}{2} \right) + \psi_{cf} E_{cf} \varepsilon_{cf,3} A_{cf} \left(h - \frac{x}{2} \right) \quad (43)$$

As $\xi > \xi_b$, the strain of concrete in the compression zone up to ultimate compressive strain, based on Fig. 20,

$$\alpha_1 k_f c b x = f_y A_s + \psi_{cf} E_{cf} A_{cf} \varepsilon_{cf,3} \quad (44)$$

$$\frac{\varepsilon_{cu}}{x_c} = \frac{\varepsilon_{cf,u}}{h - x_c} \quad (45)$$

$$x = \beta_1 x_c \quad (46)$$

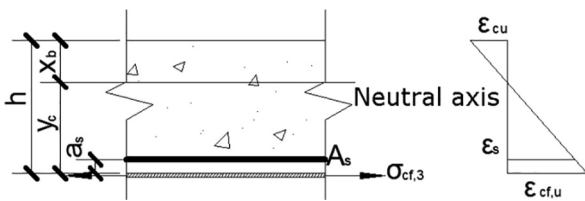


Fig. 20. Stress and strain of beams at balanced failure mode.

$$\varepsilon'_{cf,3} = \varepsilon_{cf,1} + \varepsilon'_{cf,u} = \varepsilon_{cf,0} + \varepsilon_{c,0} + \varepsilon'_{cf,u} \quad (47)$$

where $\varepsilon'_{cf,3}$ is the total strain of CFRP, $\varepsilon'_{cf,u}$ is the strain increasement of CFRP from decompression stage to ultimate stage, ε_{cu} is the ultimate compressive strain of concrete, which can be calculated by Eq. (2). The value of α_1 and β_1 are 1.0 and 0.8, respectively. The ultimate bending moment:

$$M_u = \psi f_y A_s \left(h_0 - \frac{x}{2} \right) + \psi_{cf} E_{cf} \varepsilon'_{cf,3} A_{cf} \left(h - \frac{x}{2} \right) \quad (48)$$

4.3. Comparison and analyzes of calculated and experimental results

Based on Eqs. (19), (36) (43) and (48), the cracking load, yield load and ultimate load of the specimens failed by CFRP fracture are calculated and listed in Table 4, and the calculated results with the experimental values summary as shown in Fig. 21.

As can be seen from Fig. 21, the calculated values of cracking load agree well with the test results. However, in the calculation process of yield load, the effect of compression reinforcement is not considered, and the influence of relative slip of the interface between CFRP and concrete is not considered, that is, the influence of crack expansion on the load is ignored, resulting in a smaller calculation value of yield load compared to the experimental value. As regarding to the ultimate load, the calculation value of ultimate load is generally lower than the experimental value because the hardening stage after yielding is not considered. As can be seen from Table 4, for specimens SRCU-15-90, the relative error of cracking load is 20.0% and the relative error of yield load of SRC-15-0 is 16.4% which illustrated that the theoretical formulas tend to be conservative. In general, all relative errors are below 10%, as shown in Table 4, which indicates that the proposed model could conservatively predict the bearing capacity of strengthened RC beams failed by CFRP fracture.

5. Conclusions

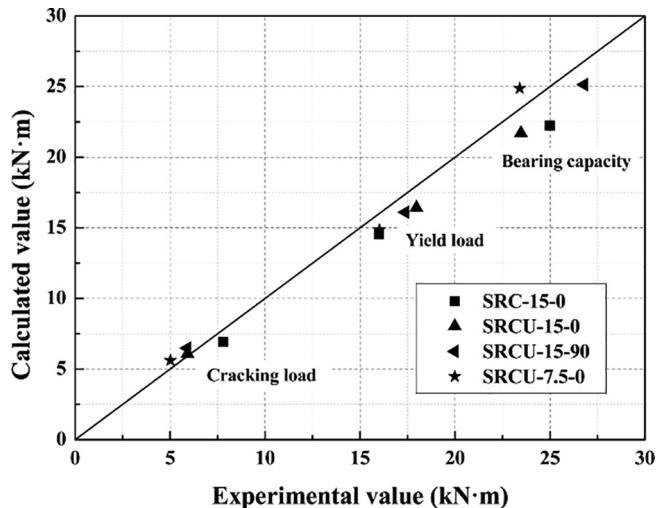
An experimental study was conducted to investigate the durability of prestressed-CFRP strengthened RC beams and the constituent materials of the strengthening system under wet-dry cycling in a chloride environment. And, a theoretical model was developed to predict the bearing capacity of RC beams strengthened with prestressed CFRP. The following conclusions can be drawn:

1. After a 90-day exposure in the laboratory-accelerated chloride environment that involved cyclic wet/dry conditions, the compressive strength and modulus of concrete increased due to the further hydration of cement, while the tensile strength of epoxy adhesive, steel bar and CFRP sheet were reduced by approximately 5–10%.
2. The exposure in chloride-containing environments generally increased the prestress losses of CFRP in the strengthened RC beams. The prestress losses of CFRP in the strengthened RC beams gradually stabilized after 30 days in room temperature, and a prestress loss in post-releasing stage around 15% was recorded for the strengthened RC beams after a 60-day exposure.
3. A 90-day exposure in the chloride environment simulating the subtropical marine climate of South China could not cause a significant adverse influence on the flexural performance of the strengthened RC beams, validating the suitability of prestressed CFRP in the strengthening of RC beams exposed to such environments.
4. CFRP strengthened beams without end anchorage were prone to CFRP debonding failure. However, as the prestressed load of CFRP increased, CFRP rupture failure might occur, and the addition of inclined U-jackets could change the failure mode of the strengthened beam from debonding failure to CFRP rupture. Additionally, the failure mode of the strengthened beams with prestressed CFRP

Table 4

Comparison of experimental and calculated results.

Specimens	Cracking load (kN·m)			Yield load (kN·m)			Ultimate load (kN·m)		
	Experimental value	Calculated value	Relative error	Experimental value	Calculated value	Relative error	Experimental value	Calculated value	Relative error
SRC-15-0	7.1	6.6	7.6%	17.0	14.6	16.4%	23.1	22.2	4.1%
SRCU-15-0	6.0	6.4	6.3%	17.8	16.4	8.5%	23.5	21.7	8.3%
SRCU-7.5-0	5.8	5.8	0.0%	16.0	14.9	7.4%	23.4	24.9	6.0%
SRCU-15-90	7.8	6.1	27.9%	17.4	16.1	8.1%	26.8	25.1	6.8%

**Fig. 21.** Experimental and calculated results.

may be changed by the exposure environment, i.e. interface debonding prior to CFRP rupture, but the debonding failure can be also prevented by inclined U-jackets.

- The proposed model based on the consideration of environmental reduction coefficient of the constituent materials is conservative for the prestressed-CFRP strengthened RC beams after a short-term exposure, which could predict the cracking, yielding and ultimate loads of strengthened RC beams failed by CFRP fracture.

CRediT authorship contribution statement

Zhongyu Lu: Writing - original draft, Validation, Resources. **Jianglin Li:** Formal analysis, Investigation, Data curation, Writing - original draft. **Jianhe Xie:** Conceptualization, Methodology, Resources, Writing - review & editing, Supervision, Funding acquisition. **Peiyan Huang:** Writing - review & editing, Investigation, Supervision, Funding acquisition. **Lingfeng Xue:** Investigation, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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